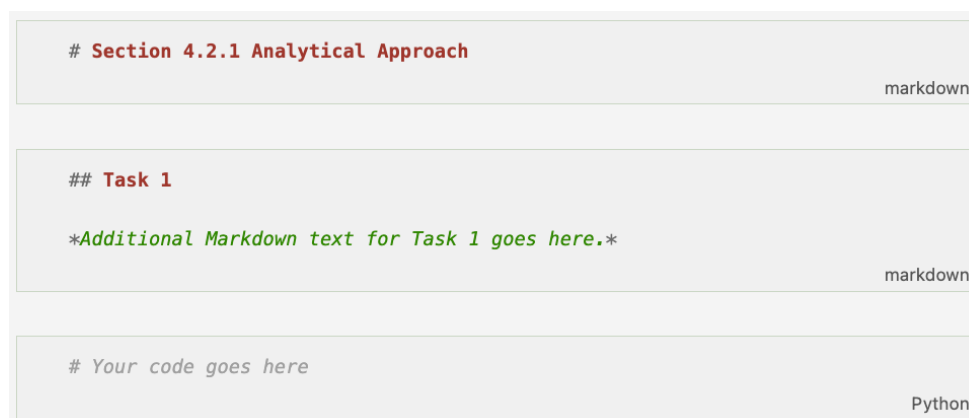


1 Instructions

1. This document outlines the tasks that you need to complete for this assessment item. Carefully read all sections of this document before attempting the assessment task.
2. This document must be read in conjunction with the general instructions and guidelines stated on the [assessment task Canvas page](#).
3. The UTS and Faculty of Science [assessment procedures and advice](#) also apply to this assessment task, so make sure you are familiar with these guidelines. The linked page provides general information about late submission penalties and special consideration.
4. Your submission will consist of (1) a single Jupyter notebook (.ipynb file) containing all implementations and responses (including discussions and mathematical derivations) related to the questions below and (2) an HTML (.html) version of the Jupyter notebook.
5. **Submission format:** While you are free to prepare your Jupyter notebook as you see fit, your submission must observe the following:
 - In the first cell of the notebook, write your name and student ID number.
 - Use Markdown section headers to label your work according to the section number and task number. For example:



- Your codes must be sufficiently annotated so that any other person familiar with the problem (other than you) can quickly understand what is happening in each section of the code and identify those sections where changes need to be made if the problem is varied.
 - Your notebook must also **display only the output requested in each task**. Suppress all other outputs. When reporting numerical quantities, round off your answers to four decimal places.
6. All submissions must be made via Canvas. Submissions made using other means (including e-mail) will not be accepted.
 7. If you have any questions about the assessment, please post these to the subject discussion forum on Canvas. If the enquiry relates to a personal matter, please contact the subject coordinator via e-mail.

2 Overview

Today is **15 August 2025** and suppose you hold a portfolio consisting of:

- a long position on an at-the-money call option on the Apple (AAPL) stock which matures in 3 months
- a short position on η_1 units of AAPL stock
- a long position on an at-the-money call option on IBM (IBM) stock which matures in 6 months
- a short position on η_2 units of the IBM stock

Historical time series of daily closing prices of AAPL and IBM from 31 December 1999 to 15 August 2025 are provided in `AT2StockPriceData.csv`.

*The objective of this assessment task is to calculate the **10-day** value-at-risk (VaR) and expected shortfall (ES) of the portfolio using various methods and modelling assumptions.*

3 Assumptions and Notation

This section enumerates the assumptions that apply throughout this assessment task. Wherever applicable, you must use the notation defined in this section and the notation that may be introduced in subsequent sections. Notation you introduce needs to be properly defined. For reference, you may also use the notation used in the lecture notes in class wherever applicable.

1. The daily observations in the provided data are equally spaced. Assume that there are 250 trading days in a year.
2. The risk management time horizon has length $\Delta t = 10/250$ corresponding to 10 trading days expressed in years.
3. Neither stock pays dividends.
4. The continuously compounded risk-free interest rate, denoted by r , is constant at 3.5% per annum for any maturity.
5. Suppose the Black-Scholes-Merton model assumptions hold. Let $S_1(t)$ and $S_2(t)$ denote the current price of the AAPL and IBM stocks, respectively. We assume that $\{S_i(t)\}_{t \geq 0}$ satisfy the stochastic differential equation

$$dS_i(t) = rS_i(t) dt + \sigma_i S_i(t) dW_i(t), \quad t \geq 0, \quad S_i(0) > 0, \quad i = 1, 2$$

under the risk-neutral probability measure $\mathbb{Q}$, where W_1 and W_2 are (possibly correlated) standard $\mathbb{Q}$ -Brownian motions and σ_1 and σ_2 denote the (constant) annualized volatility of the stock price processes.

6. Let $C_{i,t}$ denote the current price of the European call option on the Apple stock ($i = 1$) and on the IBM stock ($i = 2$). Denote by $\Delta_{i,t}$, $\Gamma_{i,t}$, and $\Theta_{i,t}$ the current value of the option delta, gamma, and theta, respectively. Assume that these quantities can be computed using the Black-Scholes-Merton equations (see Appendix A for the formula for the option theta under the Black-Scholes-Merton model).
7. Let $Z_i(t) = \ln S_i(t)$ denote the log-price of the i th stock, $i = 1, 2$. The log-prices will be used as the risk factors of the portfolio.

Using the time series notation introduced in the lectures, $Z_{i,t} = Z_i(t)$ denotes the current log-price of the i th stock and $Z_{i,t+1} = Z_i(t + \Delta t)$ denotes the log-price of the stock at time $t + \Delta t$ (i.e. after 10 trading days).

Let $X_{i,t+1} = Z_{i,t+1} - Z_{i,t}$ be the random variable denoting the log-return on the i th stock over the risk management time horizon for $i = 1, 2$. Let $\mathbf{X}_{t+1} = (X_{1,t+1}, X_{2,t+1})^\top$ denote the vector of risk factor changes.

4 Tasks

4.1 Data Preparation and Risk Factor Mapping

1. Using the daily stock prices, construct a time series of *overlapping* 10-day log-returns for each stock (note that this may induce serial correlations in the resulting time series, but using non-overlapping 10-day returns severely limits the amount of available data). Report the mean log-return for each stock. Using all available log-returns, calculate and report the covariance matrix of the log-returns using the exponentially weighted moving average (EWMA) method with smoothing constant 0.04 as of today.
2. Using the standard deviations contained in the EWMA covariance matrix estimate for the 10-day log-returns, calculate the estimated annualized volatility, which we denote by $\hat{\sigma}_i$. Report the annualized estimated volatilities.
3. Using the given information on the call options, the current stock prices, and the estimated annualized volatilities, calculate the (Black-Scholes-Merton) option price, delta, gamma, and theta. Report all quantities.
4. Let $V(t, z_1, z_2)$ denote the value of the portfolio written as a function of time t and the (current) values $z_1 = Z_{1,t}$ and $z_2 = Z_{2,t}$ of the risk factors. Let L_{t+1} denote the loss on the portfolio over the risk management time horizon $[t, t + \Delta t]$,

$$L_{t+1} := -(V_{t+1} - V_t) = -(V(t + \Delta t, Z_{1,t+1}, Z_{2,t+1}) - V(t, Z_{1,t}, Z_{2,t})).$$

Derive a delta-gamma loss approximation $L_{t+1}^{\Delta, \Gamma}$ for the portfolio *which includes the effect of time* (i.e. the loss approximation must involve the term $(\partial V / \partial t) \Delta t$) in terms of $Z_{1,t}$, $Z_{2,t}$, $X_{1,t+1}$, $X_{2,t+1}$, η_1 , η_2 , the option prices, and the option greeks. Consider only the first-order sensitivities with respect to time and the risk factors and the second-order sensitivities with respect to the risk factors.

Express the delta-gamma loss approximation in the form

$$L_{t+1}^{\Delta, \Gamma} = -c_t - \mathbf{b}_t^\top \mathbf{X}_{t+1} - \frac{1}{2} \mathbf{X}_{t+1}^\top \mathbf{B}_t \mathbf{X}_{t+1}$$

where c_t , $\mathbf{b}_t$, and $\mathbf{B}_t$ are a constant, vector, and matrix, respectively, which depend on the quantities mentioned in the previous sentence.

5. Using the current risk factor values, calculate and report the value of c_t , $\mathbf{b}_t$ and $\mathbf{B}_t$.

4.2 Market Risk Modelling

In this section, assume that the portfolio is short in one share of each stock, i.e. $\eta_1 = \eta_2 = 1$. In the sections that follow, we shall calculate the 10-day 99% VaR and ES of the portfolio using different methods and model assumptions.

4.2.1 Analytical Approach

We use the delta loss approximation $L_{t+1}^{\Delta} = -c_t - \mathbf{b}_t^\top \mathbf{X}_{t+1}$ and assume $\mathbf{X}_{t+1} \sim N_2(\hat{\boldsymbol{\mu}}_t, \hat{\boldsymbol{\Sigma}}_t)$, where $\hat{\boldsymbol{\mu}}_t$ is the mean vector and $\hat{\boldsymbol{\Sigma}}_t$ is the covariance matrix of the 10-day log-returns estimated in Task 1 in Section 4.1.

1. Calculate and report the 10-day 99% VaR and ES of the portfolio using the analytical approach.
2. State the interpretation of the figures you calculated in the previous task.

4.2.2 Historical Simulation Approach

The time series data set of historical 10-day log-returns serve as the collection of possible scenarios for the random 10-day log-return over the risk management time horizon. When the age-weighted historical simulation approach is to be used, assume an age decay parameter of 0.94.

1. Using the standard historical simulation approach and the delta-gamma loss approximation, calculate and report the 10-day 99% VaR and ES of the portfolio.
Compare your results to those obtained using the standard historical simulation and the full valuation approach. Explain why the results may differ.
2. Conduct the same analysis in the previous task but using the age-weighted historical simulation approach. Report the risk measures obtained and compare the results obtained using the loss approximation and the full valuation approach.
3. Compare and discuss the risk measures obtained through the standard and age-weighted historical simulation approaches.

4.2.3 Monte Carlo Simulation Approach

In this approach, we consider two models for the distribution of the risk factor change $\mathbf{X}_{t+1}$:

- bivariate Gaussian distribution, $\mathbf{X}_{t+1} \sim N_2(\hat{\boldsymbol{\mu}}_t, \hat{\boldsymbol{\Sigma}}_t)$ (same assumption as in the analytical approach)
- $X_{1,t+1}$ and $X_{2,t+1}$ have marginal Gaussian distributions (as implied by the lognormality assumption of underlying asset prices in the Black-Scholes-Model), but the joint distribution is modelled using a bivariate Student's t copula. The (marginal) mean and standard deviation of $X_{i,t+1}$ are given by the corresponding entries in $\hat{\boldsymbol{\mu}}_t$ and $\hat{\boldsymbol{\Sigma}}_t$ estimated in Task 1 in Section 4.1.

In this section, we fit a Student's t copula to the log-returns then calculate the required risk measures using Monte Carlo simulation.

1. Construct pseudo-observations using the empirical cumulative distribution function of the log-returns (i.e. using ranks). Using maximum likelihood estimation (MLE), estimate the parameters (the degrees of freedom $\nu > 0$ and the parameter $\rho \in (-1, 1)$ of the scale matrix) of the bivariate Student's t copula. You must write a code defining the (negative) log-likelihood function manually and perform the minimization step using `scipy.optimize.minimize`. Report your parameter estimates.
2. Using the `pycop` library, calculate the coefficients of upper and lower tail dependence of the log-returns. Use a 0.99 and 0.01 cutoff for the upper and lower tail dependence coefficients, respectively. Using Proposition 4.17 in the lecture slides on copulas and dependence, calculate the coefficient of upper and lower tail dependence implied by the bivariate t copula. Report the coefficients of tail dependence and briefly compare the results obtained. Construct a scatter plot of the pseudo-observations calculated in the previous task and discuss whether the tail dependence coefficients are consistent with the historical data.
3. **Monte Carlo VaR/ES with the Student's t copula:** Simulate 10,000 draws from the bivariate Student's t copula and use the marginal distributions of the log-returns to transform the simulated values into simulated log-returns. Use the delta-gamma loss approximation to generate simulated losses. Calculate and report the 10-day 99% VaR and ES from the empirical distribution of simulated losses.

In addition, use a full valuation approach to generate loss scenarios and report the required risk measures obtained using these loss scenarios. Compare your results to those obtained using the delta-gamma loss approximation and briefly discuss your findings.

4. **Monte Carlo VaR/ES with the bivariate Gaussian distribution:** Repeat the previous task, but now use the bivariate Gaussian distribution for the distribution of the log-returns over the risk management time horizon. Report all required risk measures.
5. Compare the risk measures obtained under the t copula and joint Gaussian distribution assumptions. Discuss why the results may be different and which joint model may be more appropriate.

4.3 Stressed VaR and ES Calculations

For the purposes of calculating stressed 10-day 99% VaR and ES estimates, log-returns observed during two time periods below shall be used as financial stress period data:

- Stress Period 1: 3 December 2007 to 30 June 2009 capturing financial stresses during the 2007-2009 Great Recession in the United States of America
- Stress Period 2: 30 January 2020 to 1 February 2021 covering approximately 12 months of financial turbulence brought about by the COVID-19 pandemic.

The following risk models will be used in calculating stressed risk measures. Additional assumptions are stated below.

- Analytical approach
- Standard historical simulation approach with full valuation
- Monte Carlo simulation approach with full valuation under the Student's t copula
- Monte Carlo simulation approach with full valuation under the bivariate Gaussian distribution.

Note that the exposure coefficients (calculated in Task 5 of Section 4.1) and the option features (including the volatility used to value the options) should not change since these quantities must reflect current market conditions (i.e. current risk factor values).

The tasks to be completed are enumerated below.

1. Compute the required risk measures using the analytical approach under the assumption that the mean and covariance of the log-returns are equal to the sample mean and sample covariance matrix (not the one calculated using the EWMA approach) of the log-returns during the stress periods.
2. Using the standard historical simulation approach with full valuation, calculate the required risk measures using the stress period log-returns as the potential future value of the log-returns.
3. To implement the Monte Carlo simulation approach, estimate the parameters of the marginal distributions of the log-returns and the t copula using the stress period log-returns. Use the same MLE approach discussed in Task 1 of Section 4.2.3. Simulate 10,000 log-returns from this model for the log-returns to calculate the required risk measures (see Task 3 of Section 4.2.3).
4. Carry out the same task as above but with a bivariate Gaussian distribution to model the distribution of the log-returns (see also Task 4 of Section 4.2.3 for reference).
5. The results from the tasks in this section must be summarized in tabular format shown below (either as a data frame object or a Markdown table).

Compare the risk measures you obtained using the stress period data and the risk measures you obtained in Section 4.2. Is it then appropriate to use the risk measures summarized in Table 1 as stressed risk measure estimates? Discuss why or why not.

Table 1: Stressed 10-day 99% VaR and ES using log-returns during Stress Periods 1 and 2.

Method	Stress Period 1		Stress Period 2	
	VaR	ES	VaR	ES
Analytical				
Standard Historical Simulation				
Monte Carlo Simulation (t Copula)				
Monte Carlo Simulation (Gaussian)				

4.4 A Simple Risk Minimization Strategy

Assume that the numbers of shares $\eta_1 > 0$ and $\eta_2 > 0$ held in a short position in the portfolio are unspecified. In this section, we are interested in determining the number of shares that are required to minimize the risk of the portfolio.

1. Using the delta loss approximation obtained in Task 4 of Section 4.1, determine analytically the number of shares that we must hold in the portfolio to minimize the risk of loss over the risk management time horizon. Justify and briefly discuss the implications of your answer.
2. Construct a function in Python which takes as input the number of shares η_1 and η_2 and returns the 10-day 99% VaR under the analytical approach (see Section 4.2.1). Assuming all stock prices and option prices and greeks remain the same, numerically minimize the analytical 10-day 99% VaR with respect to η_1 and η_2 . Ensure that η_1 and η_2 are constrained such that they assume positive values only. Discuss whether your answer here is consistent with your answer in the previous task.
3. Perform the previous task with the standard historical simulation method with a full valuation approach. Compare your findings to the result from the previous task and explain why your answers may be different.
4. Finally, perform the same task as Task 2 in this section using the Monte Carlo simulation method with the full valuation approach assuming that the joint distribution of the log-returns follow the t copula (see Task 3 in Section 4.2.3). Compare your findings to the result from the two previous tasks and explain why your answers may be different.
5. Under the optimal number of shares, what then accounts for the riskiness of the portfolio over the risk management time horizon? Discuss your answer briefly.

4.5 Summary and Conclusions

1. Display all the risk measure estimates obtained in Section 4.2 in tabular format. You are free to design table as long as it facilitates the comparison of VaR and ES estimates across the different methods. Your table can either be a displayed data frame object or a Markdown table.
2. Given the nature of the portfolio, which method do you think is most appropriate to model the portfolio's market risk? Discuss the strengths and the limitations of the method you have chosen.
3. Discuss the strengths and limitations of the stress testing methodology used in Section 4.3. What insights can be drawn from the analysis in that section?
4. Discuss the strengths and limitations of the simple risk minimization strategy you explored in Section 4.4. How do you think this approach can be generalized to more complex financial portfolios?

A Black-Scholes-Merton Formula for the Option Theta

Let $C(t, S)$ denote the Black-Scholes-Merton price at time t for the European call option with strike price K and maturity date T , where S is the current price of the underlying asset. Assuming a constant risk-free rate r , dividend yield q , and annual volatility σ , the call option theta, $\Theta(t, S) := \partial C(t, S)/\partial t$, is given by

$$\Theta(t, S) = -\frac{S\varphi(d_1(t, S))\sigma e^{-q(T-t)}}{2\sqrt{T-t}} - rKe^{-r(T-t)}\Phi(d_2(t, S)) + qS\Phi(d_1(t, S))e^{-q(T-t)},$$

where φ and Φ denote the pdf and cdf of the standard normal distribution and

$$\begin{aligned} d_1(t, S) &= \frac{1}{\sigma\sqrt{T-t}} \left[\ln\left(\frac{S}{K}\right) + \left(r - q + \frac{1}{2}\sigma^2\right)(T-t) \right] \\ d_2(t, S) &= d_1(t, S) - \sigma\sqrt{T-t}. \end{aligned}$$